

CAIT
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Abstract class and Interface

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Abstract class

```
class Shape
{
   void draw(); //not supported in java
}
```

Abstract method is the method, which contains abstract keyword on the method.

If a class contains any abstract method - then it is mandatory to add the abstract keyword to the class - abstract class.

Abstract class contains abstract method.

```
abstract class Shape {
    public abstract void draw();
}
```

When one class is extending the Shape class, we are enforcing the sub class to override the abstract method draw Or we have to make that sub class as abstract.

```
class Circle extends Shape {
    public void draw() {
        System.out.println("Circle");
    }
}
```

In java , it is not allowed to create the object for the abstract class.

Why Abstract class ?

When one class is extending the Abstract class, abstract class is enforcing the rule that Sub class should override the abstract method or make that sub class as abstract.

Interface

```
abstract class Shape
{
  public abstract void draw();
  public abstract void paint();
}
```

```
interface Shape
{
   void draw(); //public abstract is added by default
   int BONUS_PERCENTAGE = 10; //public static final --
constant variable
}
```

Interface

Feature	Abstract Class	Interface
Abstraction(before java8)	Partial abstraction	Full abstraction *(traditionally)*
Abstract methods	Allowed	Allowed
Concrete methods(before java 8)	Allowed	Not allowed Before Java 8
Concrete methods(from java 8)	Allowed	From java 8 – default and static concrete method is allowed inside the interface

Concrete method - method with definition

Interface

Without affecting the implementation classes, we can add the new feature to the interface using default and static concrete method

Default concrete method of interface - inherited to the implementation class.

Static concrete method of the interface – not inherited to the Implementation class.

Interface

To access default concrete method of the interface
Create the object for the implementation class.

To access the static method of the interface
Use interface name

THANK YOU